# Yuyuan Han

Computational imaging & robust image-free sensing

hanyuyuan6@gmail.com · yuyuan.han@mail.polimi.it · Milan, Italy
hanyuyuan6.github.io · GitHub / Hanyuyuan6 · Google Scholar · ORCID / 0009-0005-7643-6331

PhD student working on computational imaging and robust image-free sensing, from optical measurement design and reconstruction to networks that infer directly from measurements.

## Education & Research Appointments

**Xi'an Jiaotong University** · PhD student — 2022–present
Electronic Science and Technology · Master's-to-PhD track · Advisor: Huaibin Zheng

**Politecnico di Milano** · Visiting PhD student — Oct 2026–present
Department of Physics · Supervisor: Antonio Pifferi
Research focus: time-domain diffuse optical tomography for 3D brain imaging

**Northwest University** · B.Eng. — 2018–2022
Electronic Information Science and Technology

## Selected Research

### The Lift Spectrum · Robust image-free sensing — Preprint

arXiv · 2026 · Paper · Code

- Image-free single-pixel segmentation can outperform reconstruction-first inference under measurement noise with clean-trained models; introduced a content-adaptive 1D-to-2D lift and task-prioritized loss scheduling.
- 3.1–21.7 percentage-point foreground mIoU gains at 3.13% sampling and 20 dB Gaussian measurement noise across three simulated datasets; optical-bench proof of concept.

### Sparse Hadamard single-pixel imaging

Optics Express · 2025 · Paper · Code

- Tunable sparse Hadamard patterns retain orthogonality and improve noise resilience, with two training-free modes and a sparsity-dependent noise model tested in simulations and through-scattering optical experiments.

### LDGI · Ghost imaging for optical phased-array line patterns

Optics Communications · 2025 · Paper · Code

- Loop differential ghost imaging reduces spatial noise asymmetry for 1D optical phased-array line modulation; validated with a noise-symmetry metric, numerical simulations and spatial-light-modulator-generated line patterns.

## Current Research

Time-domain diffuse optical tomography for 3D brain imaging; acquisition-aware single-pixel sensing on an optical bench; exploratory question answering from measurements with a frozen vision–language model.

## Open Source & Tools

**Open source** · remote-gpu-trainer, an agent skill for Claude Code and Codex that runs, verifies and delivers GPU experiments
**Tools & lab** · PyTorch, MATLAB, COMSOL; DMD/SLM illumination, single-pixel detection, optical alignment and system integration
**Device modeling** · Geometry optimization of electro-optic phased arrays for beam steering

## Award & Service

**2024** · Second Prize, National Post-Graduate Mathematical Contest in Modeling · Code
**Reviewer** · IEEE TIP, IEEE TCI, Optics Express, Optics Letters (13 reviews, 2025–2026)